Test Plan Document

Event Management System — SE Mini Project

Version 2.0 — supersedes the Sprint 1 Test-Case Document

4 November 2026

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1. Introduction

1.1 Purpose

This Test Plan defines the test strategy, environment, pass/fail criteria, and the full set of test cases — functional, non-functional, and security — for the Event Management System (EMS). It supersedes the Sprint 1 Documents/TEST-CASE_DOCUMENT.pdf by placing the same 60 functional test cases into an IEEE-style test-plan structure, adding explicit Non-Functional Requirement coverage and a dedicated security-validation section, and recording which cases now run automatically in CI versus which remain manual or pending an environment this project does not yet have (e.g. a staging deployment for a backup/restore drill).

1.2 Scope

Covers all 10 Functional Requirements (FR-001 – FR-010) and all 7 Non-Functional Requirements (NFR-001 – NFR-007) defined in Documents/SRS_DOCUMENT.pdf, plus the security controls defined in Documents/SECURITY_DOCUMENT.pdf (SRSec). It does not cover load/performance testing beyond what the automated suite already exercises (Section 4.1), nor infrastructure-level drills (uptime monitoring, backup restoration) that require a deployed staging environment.

1.3 References

- Documents/SRS_DOCUMENT.pdf — Software Requirements Specification
- Documents/SECURITY_DOCUMENT.pdf — Security Requirements Specification (SRSec)
- Documents/SAD_DOCUMENT.pdf — Software Architecture & Design Specification
- tests/*.test.ts — the automated test suite this plan cross-references
- .github/workflows/ci.yml — CI pipeline that runs the automated subset of this plan on every push/PR to main

2. Test Items

The items under test are the ten functional areas below, each mapped to its SRS requirement and its primary implementing module (see Documents/SAD_DOCUMENT.pdf §2.4 for the full architectural traceability).

Item	FR(s)	Primary module(s)
Account signup & email verification	FR-001	app/api/auth/signup, verify
Login & password reset	FR-002	app/api/auth/login, forgot-password, reset-password
RBAC, profile, admin user management	FR-003	lib/auth.ts, app/api/admin/users
Event creation & editing	FR-004	app/api/events, lib/registration.ts (updateEventDetails)
Event cancellation & notifications	FR-005	app/api/events/[id] (PATCH), lib/registration.ts (cancelEvent, notifyEventParticipants)
Event discovery, search, filter, categories	FR-006	lib/eventFilters.ts, lib/categories.ts
Event registration & duplicate prevention	FR-007	app/api/events/[id]/register, lib/registration.ts
Capacity check & concurrency control	FR-008	lib/registration.ts (withEventLock, SERIALIZABLE tx)
Waitlist, self-cancellation, reminders	FR-009	lib/registration.ts (waitlist, promotion, reminders)
Reporting, export, statistics	FR-010	lib/export.ts, lib/stats.ts

3. Test Approach

3.1 Strategy and Levels

Three levels of testing are used throughout, following the naming convention already established in Sprint 1:

Level	ID prefix	What it exercises	Tooling
Unit	UT_	A single function or component in isolation (e.g. capacity arithmetic, a validation schema).	node:test, in-memory mock Prisma client
Integration	IT_	Two or more modules together — typically one API route plus the business-logic function it calls plus a real database.	node:test against a real PostgreSQL instance (CI service container)
System	ST_	A full user-facing flow end to end, across multiple requests/roles.	node:test driving the Next.js route handlers directly, or

Level	ID prefix	What it exercises	Tooling
			manual walkthrough where noted

Concurrency-sensitive cases (FR-008, FR-009a) are run against a real PostgreSQL SERIALIZABLE transaction, not a mock, because the correctness property under test — exactly one of N concurrent requests wins — is exactly the guarantee an in-memory mock cannot reproduce (see Documents/SAD_DOCUMENT.pdf §3.4).

3.2 Features to Be Tested

- All 10 FRs in Section 2, including every acceptance-criterion branch (success and failure paths) listed in the SRS.
- NFR-001 (performance) and NFR-002 (scalability/concurrency) at the scale the automated suite can run in CI.
- The security controls in SRSec §2–§5 that are implemented (Section 5.1).

3.3 Features Not to Be Tested

- Load at the SRS's full target scale (500 concurrent users, 10,000 total users) — the automated suite validates correctness at a smaller concurrency (10–100 simultaneous requests); full-scale load testing needs a deployed environment and a dedicated load-testing tool (e.g. k6), tracked under NFR-001 (issue #56).
- Uptime/backup-restore drills (NFR-005) and a WCAG 2.1 AA accessibility audit (NFR-006) — both require infrastructure or tooling (a staging deployment, an accessibility scanner) this project does not yet have provisioned; tracked under issues #60 and #61.
- Third-party dependencies (Postgres, SMTP provider, Vercel) are treated as trusted and are not re-tested; only EMS's own use of them is.

4. Test Environment and Pass/Fail Criteria

4.1 Environment

Element	Detail
Runtime	Node.js, TypeScript (tsx --test), Next.js 14 App Router
Database	Real PostgreSQL (via a GitHub Actions service container in CI; a local Docker Postgres in development) — not a mock, for the reasons in §3.1
Mail	nodemailer jsonTransport in NODE_ENV=test — no real network send; captured and asserted on instead
CI	GitHub Actions (.github/workflows/ci.yml), triggered on every push and pull request to main: typecheck, the full test suite, a coverage report, and a production build

Element	Detail
Current status (4 Nov 2026)	193 automated tests, 0 failing; TEST-CASE_DOCUMENT.pdf's 60 manually-run cases below are additionally re-verified against the live app where marked 'Pass'

4.2 Item Pass/Fail Criteria

- A test case passes when its actual result matches its expected result exactly, including the HTTP status code and error message text where one is specified.
- An automated case passing in CI on main is treated as the authoritative result; a manually-run case is recorded as Pass/Fail by whoever executed it, with the date in the project's execution log.
- The suite as a whole is considered releasable when every automated test passes and no open Security Validation case (Section 5.1) is rated Fail.

4.3 Suspension and Resumption Criteria

- Testing is suspended if the CI pipeline itself cannot run (e.g. the Postgres service container fails to start) — this blocks all IT_/ST_ cases, since they require a real database.
- Testing resumes once the blocking infrastructure issue is fixed; cases already passing are not re-run unless the fix touched code they exercise.

5. Test Cases

5.1 Security Validation

Each case traces to a SRSec section (Documents/SECURITY_DOCUMENT.pdf) and, where automated, to the test file that exercises it.

ID	SRSec §	Description	Result
SEC_01	2.1	Password hashes are bcrypt (cost 12); plain-text password never appears in the database or in logs.	Pass — code review: app/api/auth/signup, reset-password
SEC_02	2.2	Session cookie is HttpOnly, SameSite=Lax, and its HMAC signature is verified with a constant-time comparison.	Pass — code review: lib/session.ts
SEC_03	2.3	Account locks after repeated consecutive failed logins.	Fail — not implemented (tracked: issue #59, NFR-004)
SEC_04	3.1 / 3.2	A Participant calling an Organizer/Admin-only endpoint directly (no UI) receives 403, not partial data.	Pass — automated: tests/api-routes.test.ts
SEC_05	3.3	Organizer A cannot edit or cancel Organizer B's event by guessing its id (IDOR).	Pass — automated: tests/event-edit.test.ts, tests/event-notifications.test.ts
SEC_06	4.1	All traffic is served over HTTPS; HTTP is redirected.	Pass (platform-enforced by Vercel) — not independently re-verified in this environment
SEC_07	5.1	A SQL-injection payload in an event name/description field (e.g. ' OR 1=1 --) has no effect — Prisma parameterizes every query.	Pass — code review: no raw SQL anywhere in lib/ or app/api/
SEC_08	5.2	A script-tag payload in an event description is escaped, both in the rendered page and in outbound notification email.	Pass — automated: tests/event-notifications.test.ts ("...escape HTML" case)
SEC_0	5.3	A cross-site form submission cannot trigger a	Partial —

ID	SRSec §	Description	Result
9		state-changing action using the victim's session.	SameSite=Lax cookie only, no explicit anti-CSRF token (tracked: issue #59)
SEC_10	6.1 / 6.2	A forced application error is recorded in a central log with timestamp and severity, with no plaintext secrets in the log line.	Partial — errors are logged via console.error with context; not yet aggregated into a durable, queryable central log (tracked: issue #62, NFR-007)

Three of ten checks are not fully met (SEC_03, SEC_09, SEC_10) — each is carried into the backlog rather than marked Pass; see Documents/SAD_DOCUMENT.pdf §3.5 and §5 for the same gaps from the architecture side.

5.2 Functional Test Cases

60 cases total (6 per FR — 2 unit, 2 integration, 2 system), transcribed from the Sprint 1 execution run recorded in Documents/TEST-CASE_DOCUMENT.pdf and re-verified against the current main branch. Cases marked "automated" additionally run on every CI build.

FR-001 — Account Registration & Email Verification

ID	Description & Steps	Expected Result	Result
UT_01	Account is created with valid signup details. On Sign Up page, enter name/email/password (Amit Rao, amit.rao@test.com, Pass@123), submit.	Account created, shown 'pending verification'; login blocked until verified.	Pass
UT_02	Signup rejected for invalid email format. Enter name/password, malformed email (amitrao@@test), submit.	Rejected with 'invalid email format' error; no account created.	Pass
IT_01	Clicking the emailed verification link activates the account. Pending account exists with verification email sent. Open email, click link, attempt login.	Account becomes 'active'; login succeeds immediately after verification.	Pass
IT_02	An expired/invalid verification link is rejected. Link older than configured expiry (>24h). Click the expired link.	'Link expired/invalid' message shown; account remains inactive.	Pass
ST_01	End-to-end signup, verification, and first login. Sign up (Priya Nair), open email,	User completes signup, is verified, logs in with	Pass

ID	Description & Steps	Expected Result	Result
	click verification link, log in with same credentials.	dashboard shown.	
ST_02	Signup rejected when email already registered. An active account exists for amit.rao@test.com. Enter name/password with that email, submit.	Rejected with 'email already registered'; no duplicate account created.	Pass

FR-002 — Login & Password Reset

ID	Description & Steps	Expected Result	Result
UT_01	Login succeeds with correct credentials. Verified active account exists. Enter correct email/password, click Login.	Logged in, redirected to dashboard within 5 seconds.	Pass
UT_02	Login rejected with an incorrect password. Verified account exists. Enter correct email, wrong password, click Login.	Rejected with 'invalid credentials'; no session created.	Pass
IT_01	'Forgot Password' sends a reset email with a valid link. Account exists with registered email. Click Forgot Password, enter email, submit.	Time-limited password-reset email sent to the registered address.	Pass
IT_02	The reset link updates the password and invalidates the old one. Valid unexpired reset link exists. Open link, set NewPass@789, log in with old password.	Password updated; old password fails login; new password succeeds.	Pass
ST_01	Complete forgot-password-to-login journey. Request reset, open email, click link, set new password, log in with it.	User successfully logs in using the newly set password.	Pass
ST_02	An expired reset link is rejected. Link older than its expiry window (>1h). Click expired link, attempt to set new password.	'Link expired' message shown; password not changed.	Pass

FR-003 — RBAC, Profile Updates & Admin Account Management

ID	Description & Steps	Expected Result	Result
UT_01	A Participant cannot access the Organizer 'Create Event' page. Logged in as Participant. Navigate directly to the Create Event URL.	Access denied with an authorization error; page content not shown.	Pass
UT_02	A user can update their profile information. User is logged in. Go to	Profile updated; new contact number displayed on next	Pass

ID	Description & Steps	Expected Result	Result
	Profile, edit contact number (9876543210), save.	view.	
IT_01	Admin suspending a user blocks that user's login. Admin logged in; target account active. Open user management, select target, click Suspend.	User status changes to 'suspended'; subsequent login attempt is blocked.	Pass
IT_02	Admin deleting a user account removes login access. Admin logged in; target account exists. Select target, click Delete, confirm.	Account removed; login attempt with deleted credentials fails.	Pass
ST_01	Each role sees only its permitted menu items, end-to-end. Three accounts exist (one per role). Log in as each role in turn, note menu.	Each role's menu shows only its permitted actions (e.g. only Admin sees 'Manage Users').	Pass
ST_02	A suspended user sees a clear error when attempting to log in. Target account has been suspended by admin. Attempt login with its credentials.	Login rejected with 'account suspended'; no session/dashboard shown.	Pass

FR-004 — Event Creation & Update

ID	Description & Steps	Expected Result	Result
UT_01	An event is created when all required fields are valid. Organizer logged in. Create Event: Tech Fest, 2026-10-10, capacity 100, submit.	Event created and appears in the organizer's event list with all fields correct.	Pass
UT_02	Event creation fails when a required field is missing. Organizer logged in. Leave 'capacity' blank, submit.	Rejected with 'required field missing' error; no event created.	Pass
IT_01	Updating event details saves and reflects on the listing page. An organizer-owned event exists. Open event, edit venue to Auditorium B, save, view listing.	Updated venue saved and displayed immediately on the public listing.	Pass
IT_02	The capacity field only accepts a positive integer. Organizer on Create Event form. Enter capacity -5, submit.	Rejected with 'invalid capacity' error; no event created.	Pass
ST_01	Organizer creates then edits an event, changes visible to participants. Organizer and participant accounts exist. Create event, edit date, participant views listing.	Participant sees the updated date without any stale data.	Pass

ID	Description & Steps	Expected Result	Result
ST_02	A Participant cannot create an event via direct page access. Logged in as Participant. Navigate to Create Event page URL directly.	Access denied; redirected or shown an authorization error.	Pass

FR-005 — Event Cancellation & Update Notifications

ID	Description & Steps	Expected Result	Result
UT_01	Cancelling an event marks its status as cancelled. Active organizer-owned event exists. Open event, click Cancel, confirm.	Status changes to 'cancelled' and is flagged as such in the event list.	Pass
UT_02	A cancelled event rejects new registration attempts. Target event is cancelled. Participant opens it, clicks Register.	Rejected with 'event cancelled' message.	Pass
IT_01	Cancelling an event notifies all registrants. 2 participants registered. Organizer cancels; check notification log for each registrant.	Cancellation notification dispatched to every registered participant.	Pass
IT_02	The cancellation notification content includes correct event details. Cancellation notification has been sent. Open the received email.	Email content correctly names the event and states it has been cancelled.	Pass
ST_01	A registered participant receives the cancellation email promptly. Participant registered for an active event. Organizer cancels; participant checks inbox.	Participant's inbox contains the cancellation email.	Pass
ST_02	Cancelling an event with zero registrants completes without error. Event exists with no registrations. Open event, click Cancel, confirm.	Event cancelled successfully with no errors and no notifications sent.	Pass

FR-006 — Event Discovery, Search, Filtering & Category Management

ID	Description & Steps	Expected Result	Result
UT_01	Searching by exact event name returns the matching event. An event named 'Tech Fest' exists. Enter 'Tech Fest' in search, submit.	Results contain the 'Tech Fest' event and no unrelated events.	Pass
UT_02	Filtering by category returns only matching events. Events exist across multiple categories. Select category 'Music', apply filter.	Only events tagged 'Music' are shown.	Pass

ID	Description & Steps	Expected Result	Result
IT_01	An admin-added category appears in the organizer's event-creation form. Admin logged in. Add category 'Workshops'; organizer opens Create Event form.	'Workshops' appears as a selectable category.	Pass
IT_02	Filtering by a date range returns only events within that range. Events exist with varying dates. Set date filter 2026-10-01 to 2026-10-15, apply.	Only events within the specified range are displayed.	Pass
ST_01	Combined name search with category and date filters returns the correct intersection. Multiple events with overlapping/non-overlapping attributes. Apply search + category + date.	Only events matching all three criteria simultaneously are returned.	Pass
ST_02	Cancelled events are excluded from listing and search results. At least one cancelled event exists among active ones. View list; search for its name.	Cancelled event does not appear in the listing or in search results.	Pass

FR-007 — Event Registration & Duplicate Prevention

ID	Description & Steps	Expected Result	Result
UT_01	A valid registration creates a record with correct user, event, timestamp, status. Participant logged in; event has seats. Open event, click Register, confirm.	Record created with correct user/event/timestamp and status 'confirmed'.	Pass
UT_02	A duplicate registration by the same user for the same event is rejected. User already registered for the event. Open same event again, click Register.	Rejected with 'already registered'; no duplicate record created.	Pass
IT_01	A successful registration decrements the event's seat count. Event has N seats before registration (20). Note count, register, recheck.	Seat count is exactly N-1 after the registration commits.	Pass
IT_02	A new registration is retrievable in the organizer's registrant list. Organizer owns the event just registered for. Participant registers; organizer opens 'View Registrants'.	The newly registered participant appears in the registrant list.	Pass
ST_01	A participant registers, sees confirmation, and appears in the organizer's list. Both accounts exist; event has open seats. Participant registers, sees confirmation;	Confirmation shown; entry appears in the organizer's registrant list.	Pass

ID	Description & Steps	Expected Result	Result
	organizer views registrants.		
ST_02	Registering for a cancelled or non-existent event is rejected. Target event is cancelled or its ID no longer exists. Attempt to register via that page/ID.	Rejected with an appropriate error; no record created.	Pass

FR-008 — Seat Capacity Check & Concurrency Control

ID	Description & Steps	Expected Result	Result
UT_01	Registration is rejected when the event has zero available seats. Event capacity fully consumed (0 left). Open the full event, click Register.	Rejected with 'event full'; seat count remains 0.	Pass
UT_02	Registration succeeds and decrements the seat count when a seat is available. Event has exactly 1 seat remaining. Register for the event.	Registration succeeds; seat count becomes 0 immediately after.	Pass
IT_01	Only one of two simultaneous requests for the last seat is confirmed. Event has exactly 1 seat. Fire two simultaneous registration requests (users A, B).	Exactly one request confirmed; the other rejected as 'event full'.	Pass — automated: tests/registration.test.ts (FR-007/NFR-002)
IT_02	Seat capacity never goes negative under a simulated race condition. Event has exactly 1 seat. Fire 10 simultaneous registration requests, check final count.	Final seat count is 0, never negative; exactly one registration confirmed.	Pass — automated
ST_01	100 concurrent requests for a 1-seat event yield exactly one confirmation. Event has 1 seat; load-testing tool configured. Fire 100 concurrent requests.	Exactly 1 confirmation and 99 rejections/waitlist offers; no overbooking.	Pass — automated (scaled to 10 in CI; see NFR-002)
ST_02	A new participant sees 'Full' and is blocked from registering for a full event. Event has 0 seats remaining. Participant opens the full event page.	Page displays 'Event Full'; Register button disabled or leads to a waitlist offer.	Pass

FR-009 — Waitlist, Self-Cancellation & Notifications

ID	Description & Steps	Expected Result	Result
UT_01	Joining a waitlist records the user's position for a full event. Event full (0 seats); participant not yet registered/waitlisted. Open event, click 'Join Waitlist'.	User added to the waitlist, shown their queue position (e.g. position 1).	Pass
UT_02	A participant can cancel their own registration, freeing the seat. Participant has an active registration. Open 'My Registrations', cancel, confirm.	Status changes to 'cancelled'; the event's seat count increases by 1.	Pass
IT_01	A cancellation automatically promotes the first waitlisted user. Event full; at least one user waitlisted. Registered participant cancels.	First waitlisted user's status changes to 'confirmed'; they are notified.	Pass — automated: tests/registration.test.ts (FR-009a/NFR-002)
IT_02	A confirmation email is sent immediately after a successful registration. Participant registers for an event with open seats. Check inbox.	A registration confirmation email is received.	Pass
ST_01	A waitlisted user is auto-promoted and notified when a seat frees up. Event full; Participant X waitlisted, Participant Y registered. Y cancels.	X's status becomes 'confirmed'; X receives a promotion notification.	Pass — automated
ST_02	A reminder notification is sent to registered participants before the event date. Participant registered for an event starting within the reminder window (24h). Trigger reminder job.	Participant receives a reminder email referencing the correct event and time.	Pass

FR-010 — Reporting, Export & Platform Statistics

ID	Description & Steps	Expected Result	Result
UT_01	The displayed registered-participant count matches actual active registrations. Event has a known count of active registrations (15). Open event page, check count.	Displayed count reads exactly 15, matching the database.	Pass

ID	Description & Steps	Expected Result	Result
UT_02	The displayed available-seat count equals capacity minus active registrations. Capacity 100 with 15 active registrations. Check displayed seats-available figure.	Displayed figure reads exactly 85.	Pass
IT_01	The exported CSV participant list matches the on-screen registrant list. Organizer owns an event with 15 registrants. Open 'View Registrants', export CSV, open file.	CSV contains all 15 registrants matching the on-screen list exactly.	Pass — automated
IT_02	The exported PDF participant list matches the on-screen registrant list. Organizer owns an event with 15 registrants. Export PDF, open file.	PDF contains all 15 registrants matching the on-screen list exactly.	Pass — automated
ST_01	The admin statistics page shows correct totals for events, users, and registrations. Admin logged in; DB has known counts (50 events, 300 users, 900 registrations). Open Platform Statistics.	Displayed totals exactly match the known database counts.	Pass
ST_02	An organizer can view accurate event-wise registration info across multiple events. Organizer owns 3 events with 5, 10, 20 registrants. Open 'My Events', view each event's info.	Each event shows its correct, distinct count with no cross-event data mixing.	Pass — automated

5.3 Non-Functional Test Cases

ID	NFR	Description	Result
NFT_01	NFR-001	A CSV/PDF export of 5,000 registrations completes within 10 seconds.	Pass — automated: tests/registration.test.ts (FR-010a/NFR-001)
NFT_02	NFR-001	Event list/search/filter pages respond within 2s, and registration within 3s, at the SRS's 500-concurrent-user target.	Not yet executed — needs a load-testing tool against a deployed environment (issue #56)
NFT_03	NFR-002	100 simultaneous requests for a 1-seat event confirm exactly one and reject the rest.	Pass — automated (run at reduced scale,

ID	NFR	Description	Result
			10 concurrent, in CI): tests/registration .test.ts
NFT_04	NFR-002	Waitlist entries are promoted in strict FIFO order under concurrent cancellations.	Pass — automated: tests/registration .test.ts (FR-009a/NFR-002)
NFT_05	NFR-003	See Security Validation SEC_01, SEC_07 (Section 5.1).	Cross-referenced
NFT_06	NFR-004	See Security Validation SEC_02 – SEC_04, SEC_09, SEC_10 (Section 5.1).	Cross-referenced
NFT_07	NFR-005	A restore drill from the daily backup completes within 4 hours and loses no more than 24 hours of data.	Not yet executed — requires a provisioned staging environment with backups configured (issue #60)
NFT_08	NFR-006	90% of first-time users register for an event within 3 minutes without training; a WCAG 2.1 AA scan reports no failures.	Not yet executed — requires a usability study and an accessibility scanner (e.g. axe-core) (issue #61)
NFT_09	NFR-007	Automated test coverage of core business logic (registration, capacity, waitlist, notifications) is at least 80%.	Partial — CI currently reports ~78% (.github/workflows/ci.yml); gap tracked under issue #62
NFT_10	NFR-007	A forced application error appears in a central log with timestamp and severity.	Partial — see SEC_10

5.4 Traceability Matrix

Every FR and NFR in the SRS has at least one corresponding test case group above; this table is the reverse index back to the SRS for quick lookup.

SRS requirement	SRS §	Test case group
FR-001 Account creation	1	§5.2 FR-001 (6 cases)
FR-002 Login & password reset	1	§5.2 FR-002 (6 cases)
FR-003 Roles & user management	1	§5.2 FR-003 (6 cases)
FR-004 Create/edit events	1	§5.2 FR-004 (6 cases)
FR-005 Cancel events, notify	1	§5.2 FR-005 (6 cases)
FR-006 Browse/search/filter	1	§5.2 FR-006 (6 cases)
FR-007 Register for events	1	§5.2 FR-007 (6 cases)
FR-008 Capacity check	1	§5.2 FR-008 (6 cases)
FR-009 Waitlist & notifications	1	§5.2 FR-009 (6 cases)
FR-010 Counts, export, statistics	1	§5.2 FR-010 (6 cases)
NFR-001 Performance	2	§5.3 NFT_01–02
NFR-002 Scalability & concurrency	2	§5.3 NFT_03–04
NFR-003 Security — data protection	2	§5.1 SEC_01, SEC_07
NFR-004 Security — access control	2	§5.1 SEC_02–04, SEC_09, SEC_10
NFR-005 Availability & recovery	2	§5.3 NFT_07
NFR-006 Usability & accessibility	2	§5.3 NFT_08
NFR-007 Maintainability	2	§5.3 NFT_09–10

6. Execution Summary

As of 4 November 2026: 193 automated tests pass in CI (0 failing), covering every FR's integration/system paths plus the NFR-001/002 and most Section 5.1 security cases. All 60 manually-tracked functional cases in Section 5.2 are recorded Pass against the current main branch. Seven items remain genuinely open rather than quietly marked Pass: account lockout (SEC_03), explicit CSRF tokens (SEC_09), a durable central audit log (SEC_10 / NFT_10), full-scale load testing (NFT_02), a backup/restore drill (NFT_07), an accessibility audit (NFT_08), and closing the 78% → 80% coverage

gap (NFT_09). These map one-to-one onto the Sprint 3 NFR backlog items (issues #56–#62) and the Deviations section of Documents/SAD_DOCUMENT.pdf.